BlueWave Resource Partners x Mirror Factory

Project Aqueduct Product Scope Document

Prepared by: Mirror Factory
Prepared for: Charlie Lewis
Date: April 5, 2026
Revision: 0.2 (Draft)

1. Scope Summary

The Client

Company	BlueWave Resource Partners — staffing and recruiting firm, Orlando, FL
Revenue	~$40M annually
Team	<40 internal employees, 18–20 recruiters, 200+ W-2 contractors placed at client sites
Decision-Maker	Charlie Lewis, Founder/CEO
Strategic Driver	Build proprietary software IP ahead of a planned business exit in ~2 years

The Problem

BlueWave's recruiters spend 75–80% of their time on unqualified candidates. When a new job requisition arrives (e.g., SpaceX needs 30 electrical engineers in two weeks), it takes days of manual sorting through hundreds of applicants before a recruiter even makes a call. Inbound calls (~8–10/day) consume admin and recruiter time on low-value tasks like password resets and status checks. BlueWave evaluated existing solutions (RingCentral AIR, Bullhorn Amplify, Tenzo AI) and found none that solve the full problem.

The Solution

Project Aqueduct is a proprietary AI system with two core components:

- **Inbound Call Triage (MVP)** — AI receptionist answers calls, determines intent, and routes to Sales, Admin/HR, or Candidates. Candidate calls transition to a recruiter agent for pre-screening. Every call generates a progressive ticket with contextual summaries.
- **Outbound Candidate Screening (Parallel Development)** — On receipt of a new job req, the system screens hundreds of candidates against qualifying criteria and delivers a qualified shortlist within one hour.

Architecture

Two-layer model with separated IP ownership:

Layer	Owner	Contains
Aqueduct (Orchestration)	BlueWave	Routing logic, ticketing, staffing integrations, screening rubrics, messaging bus
Layers (Platform)	Mirror Factory	Context management, agent profiles, rules, integration framework, authoring tools

Aqueduct functions independently of Layers. BlueWave owns their product; Mirror Factory retains and evolves its platform.

Phased Delivery

Phase	Duration	Focus
Design & Development	4–6 weeks	Build MVP (inbound triage), begin outbound screening architecture
Pilot	~90 days	Internal pilot with progressive delegation, continuous QA, outbound development continues
Expansion	TBD	Activate outbound screening, evaluate productization, design additional tools

Key Outcomes

Outcome	Current State	Target State
Recruiter capacity	~1,000 candidates at capacity	5,000–10,000 with AI assistance
Time-to-qualified-shortlist	Days of manual sorting	Under 1 hour
Low-value call handling	Manual triage, 8–10 calls/day	AI-resolved or intelligently routed
Software IP	None — using third-party SaaS	Proprietary asset on balance sheet

2. Product Vision & Objectives

Strategic Context

BlueWave is a bootstrapped, revenue-funded staffing firm that has grown from sub-$1M in 2014 to approximately $40M annually. Charlie Lewis intends to exit the business within two years. Owning proprietary software IP — not licensing third-party tools — is a strategic lever to maximize the exit valuation multiple. Every architecture and scope decision in this engagement should be evaluated against this exit-readiness lens.

No existing product handles the full problem: multi-caller-type inbound triage with deep Bullhorn integration, combined with outbound screening at scale. This gap validates the custom-build thesis.

Jobs to Be Done

These JTBD statements capture what BlueWave is hiring this system to accomplish, derived directly from the March 31 discovery meeting. Each statement connects to specific features in Section 5.

- **When** a call comes into BlueWave's main number, **I want** the caller's intent identified and routed to the right person automatically, **so that** recruiters and admins only handle calls requiring human judgment. *(→ AI Receptionist, §5.1.1)*
- **When** we receive a new job requisition, **I want** hundreds of candidates screened against the real qualifying criteria within the first hour, **so that** my recruiters start calling qualified, interested people instead of sorting through databases for days. *(→ Outbound Screening, §5.2)*
- **When** a candidate calls in, **I want** them pre-screened by an AI recruiter that captures structured evaluation data and writes it to Bullhorn, **so that** the human recruiter who follows up already knows if this person is a fit. *(→ Recruiter Agent, §5.1.5)*
- **When** someone has called multiple times about the same issue, **I want** all of that history merged into a single context-rich ticket, **so that** whoever picks up the next call has full context and the caller doesn't have to repeat themselves. *(→ Native Ticketing, §5.3)*
- **When** an AI agent handles a call or screens a candidate, **I want** a full action trace I can audit weekly, **so that** I can catch mistakes, improve the system, and build trust progressively. *(→ Audit Interface, §5.5)*

Core Objectives

- **Eliminate low-value inbound call handling.** Route calls intelligently so recruiters, admins, and sales staff only handle calls that require human judgment. Resolve self-service tasks (e.g., ADP password resets, status checks) without human intervention.
- **Increase recruiter capacity 5–10x.** Move from approximately 1,000 candidates per recruiter at capacity to intelligently managing 5,000–10,000 through AI-assisted screening and triage.
- **Compress time-to-qualified-shortlist.** Reduce the cycle from receiving a new job requisition to having a qualified, contacted candidate shortlist from days of manual sorting and calling to under one hour.
- **Create standalone software IP.** Architect Aqueduct as a separable asset on the balance sheet — owned entirely by BlueWave — that increases company valuation at exit.
- **Build a native ticketing and context system.** Every call generates a progressive ticket with contextual summaries tailored to the recipient, enabling warm handoffs to humans with full context.

Design Philosophy

Charlie emphasized a core principle during discovery: *get humans talking to the right people faster, not replace humans.* BlueWave is a relationship-driven business. AI augments recruiter capacity and decision quality — it does not remove the human element. The system should be

modeled like managing a team of interns: given clear instructions, deployed in a controlled environment, audited regularly, and improved through feedback loops.

3. System Architecture

Two-Layer Model

Project Aqueduct is built on a two-layer architecture that separates client-specific orchestration from reusable platform infrastructure. This separation serves two purposes: it gives BlueWave clean IP ownership of their business logic, and it gives Mirror Factory a reusable platform foundation for future engagements.

Aqueduct — Orchestration Layer (BlueWave IP)

Aqueduct is the orchestration layer that sits on top of Layers. It contains all business logic, routing decisions, and staffing-specific functionality. BlueWave owns this IP outright.

Aqueduct encompasses:

- Call routing logic and decision engine (the rules governing which calls go where, under what conditions)
- Native ticketing system (progressive ticket creation, caller ID merging, contextual summaries)
- Staffing-specific integrations (Bullhorn ATS/CRM read/write, RingCentral telephony)
- Human/agent coordination workflows (escalation paths, handoff protocols, notification routing)
- Recruiter agent screening logic and rubrics (qualifying criteria, evaluation scoring, structured output)
- Internal messaging bus (SOP-driven routing across text, Slack, and email)
- Audit trail and reporting interface (agent action trace, weekly review tooling, error flagging)

Layers — Platform Layer (Mirror Factory IP)

Layers provides the context management infrastructure that Aqueduct runs on. It is not required for Aqueduct to function — Aqueduct can operate with generic markdown-based context — but Layers provides best-practice scaffolding that makes the system faster to build, easier to maintain, and higher quality.

Layers encompasses:

- Context management infrastructure (how agent context is structured, stored, and retrieved)
- Agent profiles, rules, and skills definitions (the framework for defining what agents know and can do)

- Integration framework and connectors (standardized patterns for connecting to external systems)
- Authoring tools (lightweight interfaces for viewing and editing context — comparable to Obsidian for knowledge management)
- Best practices for context structuring (analogous to Unity over raw DirectX — faster to build on, not strictly required)

Layers is portable and reusable. Any custom authoring tools developed during this engagement remain in Layers and carry forward to future Mirror Factory clients.

Relationship Between Layers

Aqueduct sits on top of Layers but is not locked in. BlueWave can unplug Layers and substitute generic markdown — the system still functions, just without the best-practice scaffolding. This ensures BlueWave's IP independence while allowing Mirror Factory to retain and evolve its platform.

Deployment Considerations

Deployment model is TBD and will be resolved during the Design & Development phase. Options discussed:

- **Cloud (e.g., AWS Bedrock):** Enterprise-grade infrastructure, managed scaling. Not fully air-gapped — tokens still touch external infrastructure. Security implications need evaluation.
- **Local models (e.g., Gemma 26B+):** Full data privacy, no external data exposure. Viable if the testing and validation pipeline is strong enough. Higher operational overhead.
- **Hybrid:** Cloud for non-sensitive operations, local for candidate data processing. Balances security with operational simplicity.

The deployment decision affects security posture, operating costs, voice model selection, and the QA approach. BlueWave's requirement is a private, closed-loop deployment with no external data exposure.

Implementation Stack

Implementation approach will be finalized during Design & Development. Candidates discussed:

- Python-based orchestrator (similar to existing Mirror Factory patterns)
- Next.js modeled on open-source agent frameworks (OpenAI/Anthropic patterns)
- Voice model TBD: cloud options (ElevenLabs, AWS Bedrock) vs. local models
- Standards for building orchestrators to be established during this engagement

---

4. Agent Architecture

This section defines the AI agent system that powers Aqueduct's core functionality. Agent architecture decisions made here affect scope, feasibility, and the testing strategy.

Agent Roster

Agent	Role	Activation Trigger	Phase
Receptionist	Answers inbound calls, gathers intent, routes to the correct path	Any inbound call to BlueWave's main number	MVP
Recruiter	Conducts pre-screening conversations, evaluates candidates against a rubric, produces structured output	Candidate call identified by Receptionist	MVP
Screening Agent	Initiates outbound contact to candidates, evaluates against job req criteria, compiles shortlists	New job requisition received in Bullhorn	Parallel Dev

Agent Design Principles

- **Single responsibility per agent.** Each agent has one clear job. The Receptionist routes; it does not screen. The Recruiter screens; it does not route.
- **Progressive delegation.** Agents start with low-stakes tasks and expand responsibility as confidence builds during the Pilot. The system begins with simple routing (Admin/status checks) before handling candidate screening calls.
- **Human escalation as default.** When an agent encounters uncertainty, the default is to escalate to a human with full context — not to guess. The threshold for escalation will be tuned during the Pilot.
- **Auditable action trace.** Every agent decision is logged: what data was accessed, what routing decision was made, what output was generated. This trace feeds the audit interface and enables the weekly review workflow.

Inter-Agent Communication

The Receptionist and Recruiter agents operate within the same call. When the Receptionist identifies a candidate, it performs a warm handoff to the Recruiter agent — the caller experiences a seamless transition, not a transfer. Context gathered by the Receptionist (caller name, intent, any Bullhorn match) is passed to the Recruiter agent so the screening conversation starts with context, not from scratch.

The Screening Agent (outbound) operates independently of the inbound agents. It is triggered by Bullhorn events (new requisition), not by calls.

Agent Context Management

Each agent's behavior is governed by its context profile within Layers:

- **System prompt architecture** — defines the agent's role, personality, boundaries, and escalation rules
- **Tool definitions** — what external systems the agent can access and what operations it can perform (e.g., Recruiter agent can read Bullhorn candidate records and write evaluation data)
- **Guardrails** — explicit boundaries on what the agent should never do (e.g., never promise a candidate they have the job, never share salary information that hasn't been authorized)
- **Versioning** — context profiles are versioned so changes can be rolled back if a new version underperforms

---

5. Feature Specifications

Features are organized by component. Each feature includes a description, acceptance criteria, phase designation, and priority (Must/Should/Could per MoSCoW). AI-specific features include a hypothesis statement with validation criteria.

5.1 Inbound Call Triage (MVP)

5.1.1 AI Receptionist Agent

Priority: Must

Description: An AI agent that answers inbound calls to BlueWave's main number, introduces itself, gathers caller intent through natural conversation, and routes the call to the appropriate department or agent based on configurable rules.

Hypothesis: We believe an LLM-powered voice agent can accurately classify caller intent into one of three routing paths (Sales, Admin/HR, Candidates) with 90%+ accuracy given a structured greeting and 2–3 clarifying questions. This will be validated during the Pilot phase with a target of [TBD] live calls.

Acceptance Criteria:

- Agent answers inbound calls within 2 rings and delivers a natural, branded greeting
- Agent determines caller intent through open-ended conversation (not menu trees)
- Agent routes calls to one of three primary paths: Sales, Admin/HR, or Candidates
- Agent resolves self-service requests without human escalation where possible (e.g., ADP password reset instructions, basic status checks)
- Agent creates a ticket for every call (see §5.3 Native Ticketing)
- Routing rules are configurable without code changes
- Agent gracefully handles edge cases: unclear intent, caller frustration, requests to speak to a human immediately

5.1.2 Sales Path Routing

Priority: Must

Description: Inbound calls identified as sales-related (prospective clients, business development inquiries, partnership requests) are routed to the appropriate sales team member or voicemail.

Acceptance Criteria:

- Calls identified as sales/business development are routed per BlueWave's Sales routing SOPs
- Caller context (name, company, stated need) is captured and included in the handoff ticket
- If no sales team member is available, the agent captures a message and notifies the appropriate person via the internal messaging bus

Dependency: Sales routing SOPs must be documented by BlueWave (see §14)

5.1.3 Admin/HR Path Routing

Priority: Must

Description: Inbound calls related to benefits questions, ADP issues, internal department routing, and general administrative inquiries are routed to the Admin/HR team or resolved by the agent directly.

Acceptance Criteria:

- Agent identifies admin/HR intent and routes per BlueWave's Admin routing SOPs
- Common self-service tasks are handled by the agent without escalation (e.g., ADP password reset instructions, benefits information from a knowledge base)
- Escalation paths are defined for issues requiring human judgment
- Caller context and resolution status are captured in the ticket

Dependency: Admin routing SOPs and self-service knowledge base content must be provided by BlueWave (see §14)

5.1.4 Candidate Path Routing

Priority: Must

Description: Inbound calls from candidates (new applicants and existing candidates checking status, scheduling interviews, or inquiring about positions) are identified, classified as new or existing, and routed accordingly.

Acceptance Criteria:

- Agent distinguishes between new candidates and existing candidates (via caller ID, name match against Bullhorn, or stated intent)
- New candidates are routed to the Recruiter Agent for pre-screening (see §5.1.5)
- Existing candidates checking status receive automated updates where data is available in Bullhorn, with escalation to a human recruiter if the query requires judgment

- All candidate interactions are logged to Bullhorn

5.1.5 Recruiter Agent (Pre-Screening)

Priority: Must

Description: When a candidate call is identified, the system transitions from the Receptionist Agent to a Recruiter Agent that leads a structured pre-screening conversation. The Recruiter Agent evaluates the candidate against a rubric, captures structured data, and produces an evaluation summary.

Hypothesis: We believe an AI agent can lead a structured pre-screening conversation that captures enough structured data (3–5 qualifying criteria evaluated, candidate availability, compensation expectations) to produce a recruiter-ready assessment that accelerates — not replaces — the human recruiter's follow-up decision. Validation: during Pilot, compare agent-screened candidates against recruiter-screened candidates for assessment agreement rate.

Acceptance Criteria:

- Recruiter Agent leads the conversation from a pre-screening perspective (agent drives, candidate responds)
- Agent evaluates candidate against 3–5 qualifying criteria defined per job type or general screening rubric
- Structured output is captured: candidate profile, evaluation scores against criteria, call summary, and recruiter-ready assessment
- Output is written to Bullhorn (candidate record, tear sheet, and/or notes)
- If the candidate is a strong match for an active requisition, the ticket is flagged for immediate recruiter follow-up
- Conversation feels natural — not like a survey or checklist

Dependency: BlueWave must define the 3–5 most common qualifying criteria used across job types (see §14)

5.2 Outbound Candidate Screening (Parallel Development)

5.2.1 Requisition Ingestion

Priority: Must (for outbound component)

Description: When BlueWave receives a new job requisition, the system ingests the job description, identifies the actual qualifying criteria (stripping out boilerplate), and prepares a screening profile.

Hypothesis: We believe an LLM can reliably extract the 3–5 critical qualifying criteria from a job description, separating them from standard boilerplate, with sufficient accuracy that recruiters only need to review and adjust rather than define from scratch. Validation: compare agent-extracted criteria against recruiter-defined criteria across 20+ historical job requisitions.

Acceptance Criteria:

- System accepts job requisitions from Bullhorn (automated) or manual input
- System parses the job description and extracts the 3–5 critical qualifying criteria
- Screening profile is generated and available for review before outbound screening begins
- Qualifying criteria can be manually adjusted by a recruiter before screening launch

5.2.2 Candidate Matching and Outreach

Priority: Must (for outbound component)

Description: The system searches BlueWave's Bullhorn database (~200K candidates), identifies potential matches against the screening profile, and initiates outreach to qualified candidates simultaneously via call, text, and/or email.

Acceptance Criteria:

- System searches Bullhorn candidate database against screening profile criteria
- Outreach is initiated to matched candidates simultaneously (target: hundreds per requisition)
- Outreach channels include phone, text, and email (channel selection based on candidate data availability and BlueWave preferences)
- Candidate responses are captured, evaluated against the screening rubric, and scored
- Qualified, interested candidates are compiled into a shortlist and pushed to a Bullhorn tear sheet
- Target: qualified shortlist delivered within 1 hour of requisition receipt
- Slack notification sent to recruiting team (e.g., "45 screened electrical engineers added to tear sheet for Job #142")

Dependency: Bullhorn API read/write confirmation (see §14). Per-click sourcing tools excluded until cost controls are established.

5.2.3 Screening Agent Learning

Priority: Could (Expansion phase)

Description: The outbound screening agent improves over time by learning from response rates, bounce rates, screening question effectiveness, and placement outcomes.

Acceptance Criteria:

- System tracks response rates per outreach channel and candidate segment
- System tracks which screening questions best predict candidate quality
- Learning insights are surfaced in the audit interface for recruiter review
- Agent behavior can be adjusted based on accumulated learning without code changes

5.3 Native Ticketing System (MVP)

5.3.1 Ticket Creation and Progressive Fill

Priority: Must

Description: Every inbound call immediately generates a blank ticket that fills progressively as the call unfolds. The ticket is not a raw transcript — it is a structured, contextual record of the interaction.

Acceptance Criteria:

- Blank ticket is created the instant a call connects, before the agent speaks
- Ticket captures: caller ID, timestamp, intent classification, routing decision, conversation summary, evaluation data (if applicable), and resolution status
- Ticket fills progressively during the call — information is added as it becomes available, not retroactively after the call ends
- Ticket includes full action trace (every agent decision is logged and traceable)

5.3.2 Repeat Caller Merging

Priority: Must

Description: When the same person calls again (identified by caller ID, name, or other matching criteria), tickets are automatically merged to create a compound interaction history.

Acceptance Criteria:

- System identifies repeat callers via phone number, name match, or Bullhorn candidate ID
- Repeat caller tickets are merged into a single interaction history
- Prior context surfaces automatically when a repeat caller is identified (e.g., "This is Fraser's 6th call about the Disney electrical engineer position")
- Merged ticket history is accessible to both agents and humans

5.3.3 Contextual Handoff Summaries

Priority: Must

Description: When a call must be routed to a human, the ticket provides a contextual summary tailored to the specific recipient — not a generic transcript dump.

Acceptance Criteria:

- Ticket summary is generated in a format appropriate to the recipient's role:
- Charlie receives escalation context (e.g., "Fraser has called 16 times — recommend a direct call, he's escalating")
- Recruiters receive candidate-fit summaries (e.g., "Strong match for Job #142 — 4/5 criteria met, available immediately, requesting $85/hr")
- Admin receives resolution context (e.g., "ADP password reset — caller attempted self-service, needs manual intervention")
- Human recipient receives the summary via the internal messaging bus before the call is transferred (where possible) or immediately after

- No re-explanation is required — the human has full context from the ticket

5.4 Internal Messaging Bus (MVP)

5.4.1 SOP-Driven Message Routing

Priority: Must

Description: A central routing layer that understands BlueWave's internal communication SOPs and delivers agent notifications, ticket summaries, and escalation alerts to the right person in the right channel.

Acceptance Criteria:

- System routes messages across three channels: text (SMS), Slack, and email
- Routing rules are configured per BlueWave's SOPs: who gets what message type, department-specific rules, and urgency tiers
- Routing rules are configurable without code changes
- Each message includes contextual information from the relevant ticket

Dependency: BlueWave must document their internal messaging SOPs (see §14)

5.4.2 Daily Tear Sheet Notifications

Priority: Should (aligned with outbound screening activation)

Description: Daily summary notifications to the recruiting team via Slack with updates on candidate activity, tear sheet additions, and screening results.

Acceptance Criteria:

- Daily digest notification sent to a configurable Slack channel at a scheduled time
- Digest includes: new candidates screened, tear sheet additions, notable interactions, and any flagged issues
- Notification links directly to relevant Bullhorn records

5.5 Audit and Reporting Interface (MVP)

5.5.1 Agent Action Audit Trail

Priority: Must

Description: Every agent action is logged and traceable. The BlueWave team can review agent behavior, flag errors, and provide feedback that improves the system.

Acceptance Criteria:

- Full action trace for every agent interaction (decision tree path, data accessed, outputs generated)
- Audit interface is CMS-level simple — usable by BlueWave's non-technical team
- Weekly review workflow is supported (review last week's interactions, flag errors, approve corrections)

5.5.2 Error Flagging and Feedback Loop

Priority: Must

Description: BlueWave team members can flag incorrect agent actions and provide corrections that are incorporated into the agent's behavior.

Acceptance Criteria:

- Any team member can flag an agent action as incorrect from the audit interface
- Flagged actions include a correction note explaining what should have happened
- Corrections are reviewed and incorporated into agent context/rules
- System tracks correction history to identify recurring error patterns

---

6. Data Requirements

Data Sources

Source	Type	Volume	Access	Phase
Bullhorn	Candidate records, job reqs, tear sheets, contact info	~200K candidates	API (read/write — access level TBD)	MVP
RingCentral	Call audio, call metadata, routing events	~8–10 calls/day inbound	API (TBD during Design)	MVP
BlueWave website job board	Candidate applications	Syncs to Bullhorn via Haley/Daily Marketing	Indirect via Bullhorn	MVP
Slack	Channel membership, notification delivery	Internal team	Standard API	MVP
Sourcing tools (LinkedIn, Indeed, Dice, Zip)	Candidate profiles, outreach responses	Excluded until cost controls established	Deferred	Expansion

Data Quality Considerations

- Bullhorn data quality varies. LinkedIn profiles may be outdated, resumes stale. The system must handle incomplete or inconsistent candidate records gracefully — not fail silently on missing fields.
- Cross-referencing candidates across source systems is the industry's biggest gap and BlueWave's core opportunity. The system should be designed to handle candidate deduplication across Bullhorn records originating from different sources.

- Call audio quality will vary: mobile calls, speakerphone, background noise. The voice pipeline must be tested against realistic audio conditions, not clean studio recordings.

Data Privacy and Ownership

- **Closed-loop deployment required.** No candidate data, call recordings, or business data should be exposed to external systems beyond explicitly integrated partners.
- **BlueWave owns all data** processed through Aqueduct. Mirror Factory has no data retention rights beyond what is needed for active development and support.
- **Cloud vs. local trade-off:** Cloud models (including AWS Bedrock) are not fully air-gapped. This trade-off must be explicitly evaluated and accepted during Design & Development. If unacceptable, local models (Gemma 26B+) are viable with a strong validation pipeline.

7. Integration Map

7.1 Bullhorn — ATS/CRM (Core)

Field	Detail
Role	Primary data store for candidate records, job requisitions, tear sheets, and placement data
Data Volume	~200,000 candidate records
Required Operations	Read: candidate records, job requisitions, tear sheets, contact information. Write: call transcripts, screening evaluations, candidate notes, tear sheet entries, agent interaction logs
Status	API access level needs confirmation from BlueWave (see §14)
Priority	Critical — scope and feasibility of the entire system depend on Bullhorn API capabilities
Known Risk	Data quality varies. Cross-referencing candidates across source systems is the core opportunity and the biggest technical challenge

7.2 RingCentral — Telephony (Core)

Field	Detail
Role	Call routing infrastructure. Inbound calls flow through RingCentral to the AI receptionist agent
Required Operations	Inbound call routing, call transfer, call recording, telephony infrastructure

Status	BlueWave currently uses RingCentral. Integration approach TBD during Design & Development
Priority	Critical — telephony is the entry point for all inbound triage
Known Context	RingCentral AIR was evaluated but lacks Bullhorn integration and data portability

7.3 Slack — Internal Communications (Required)

Field	Detail
Role	Internal notification channel for agent alerts, daily tear sheet updates, escalation notifications
Required Operations	Send messages to channels and direct messages. Read channel membership for routing
Status	BlueWave uses Slack. Integration is straightforward
Priority	Required — part of the internal messaging bus

7.4 Website Job Board (Required)

Field	Detail
Role	Candidate intake channel. Third-party hosted, syncs to Bullhorn, distributes to ~100 job boards
Required Operations	Read: candidate applications. Potential future: chatbot integration (Expansion)
Status	Currently syncs to Bullhorn automatically. Website being rebranded by Go Marketing
Priority	Required for candidate flow. Direct integration may not be needed if Bullhorn sync suffices

7.5 Sourcing Tools — LinkedIn Recruiter, ZipRecruiter, Indeed, Dice (Deferred)

Field	Detail
Role	Candidate sourcing channels used by recruiters
Status	Deferred until outbound screening component is activated
Priority	Deferred — per-click cost structures require budget guardrails before agent access
Known Risk	These platforms build technology moats around data. API access is limited

7.6 ADP — HR/Payroll (Reference Only)

Field	Detail
Role	HR and payroll system. Referenced in admin routing for password resets and benefits questions
Required Operations	None directly. Agent uses knowledge base content for ADP-related self-service
Priority	Reference only — no direct integration planned

8. User Experience Surfaces

8.1 External Voice — Caller-Facing

The external voice experience is the primary interface for all inbound callers. It must feel natural, professional, and human-like. This is the highest-risk UX surface because voice quality directly affects caller trust and willingness to engage.

Experience Goals:

- Callers should not feel like they are navigating a phone tree or talking to a robot
- Conversations should be open-ended and adaptive, not scripted menus
- The agent should handle natural conversational patterns: interruptions, corrections, pauses, and topic changes
- Voice quality (latency, naturalness, tone) must support a professional business conversation

Voice Latency Budget:

Humans notice pauses after ~300 milliseconds. Conversations feel broken above 800ms. The voice pipeline has three latency-critical components:

Component	Target Latency	Notes
Speech-to-text (STT)	100–300ms	Real-time streaming preferred over batch
LLM inference	200–500ms	Streaming token output reduces perceived latency
Text-to-speech (TTS)	100–300ms	First-audio latency more important than total generation time
Total pipeline	**<800ms**	**Target for natural conversation feel**

Turn-taking behavior (detecting when the caller has finished speaking vs. pausing mid-thought) is a critical design decision. Semantic turn detection reduces response times compared to silence-based detection but requires more sophisticated implementation. This will be evaluated during Design & Development.

Interaction Pattern:

- Call connects → Agent greets caller with a branded introduction
- Agent asks open-ended questions to determine intent
- Agent classifies intent and routes accordingly
- If routing to a human: agent provides context summary and transfers the call
- If resolving directly: agent handles the request and confirms resolution
- If transitioning to Recruiter Agent: seamless handoff within the same call

8.2 Internal Messaging — Team-Facing

The internal messaging experience is how BlueWave's team receives agent notifications, ticket summaries, and escalation alerts. It is not a chat interface — it is a routing layer that delivers the right information to the right person in the right channel.

Experience Goals:

- Team members receive information where they already work (text, Slack, or email) — no new app to check
- Message format is contextual to the recipient and the situation
- Urgency tiers are respected: urgent escalations via text, routine updates via Slack, formal communications via email
- Information density is appropriate — enough context to act, not a wall of text

Interaction Pattern:

- Agent generates a notification (new ticket, escalation, daily digest, etc.)
- Messaging bus determines recipient and channel based on SOP rules
- Message is formatted with contextual summary appropriate to the recipient's role
- Recipient can act directly from the notification (e.g., click through to Bullhorn record)

---

9. Phased Roadmap

Phase 1: Design & Development

Estimated Duration: 4–6 weeks

Target Start: Mid-April 2026

Objectives:

- Establish the design and development plan for the MVP (inbound triage)
- Define product and project milestones for subsequent phases
- Build the MVP: AI Receptionist, candidate routing with Recruiter Agent, native ticketing, and messaging bus
- Begin parallel development of outbound screening architecture
- Internal QA and prototype validation

Milestone Gate — Exit Criteria for Phase 1:

☐ Inbound call flow functional end-to-end in staging (call → receptionist → routing → ticket creation)

☐ Recruiter Agent conducts basic pre-screening and produces structured output

☐ Ticketing system creates, fills, and merges tickets correctly

☐ Bullhorn integration confirmed: read/write on candidate records and tear sheets

☐ RingCentral integration confirmed: inbound calls route to AI agent

☐ Internal messaging bus delivers notifications to Slack

☐ QA plan defined and resourced for Pilot phase

☐ Phase 2 milestones and success criteria defined

Phase 2: Pilot

Estimated Duration: ~90 days

Target Start: June 2026

Objectives:

- Internal pilot focused on inbound triage — BlueWave team uses the system before external exposure
- Progressive delegation: start with low-stakes call types, expand as confidence builds
- Continued development and refinement based on real-world usage
- Continued parallel development of outbound screening
- Continuous QA — bugs and edge cases captured and resolved in real time

Milestone Gate — Exit Criteria for Phase 2:

☐ System has handled a minimum of [TBD] live inbound calls

☐ Call resolution rate meets or exceeds target KPI (TBD — see §14)

☐ Recruiter Agent screening accuracy validated by recruiting team

☐ Error rate within acceptable threshold (defined during Phase 1)

☐ BlueWave team confidence supports full production deployment

☐ Outbound screening component architecturally complete and ready for activation

☐ Go/No-Go decision: proceed to Expansion or iterate

Phase 3: Expansion

Estimated Duration: TBD

Target Start: Q4 2026

Objectives:

- Activate outbound candidate screening component
- Establish cost controls for per-click sourcing tools before enabling agent access
- Evaluate packaging Aqueduct 1.0 for other staffing agencies (e.g., NextPath)
- Design additional tools for sales, compliance, and outbound workflows
- Implement screening agent learning and automated improvement
- Expand channel coverage: evaluate website chatbot, shared inbox management

Milestone Gate — Exit Criteria for Phase 3:

☐ Outbound screening delivers qualified shortlist within 1 hour of req receipt
☐ Cost controls verified for per-click sourcing tools
☐ Productization assessment complete (internal tool vs. licensable product decision)
☐ Additional tools scoped and prioritized for next engagement cycle

---

10. Testing & QA Strategy

The Challenge

Voice-based AI systems present unique QA challenges. Unlike text-based agents where millions of automated test instances can be run, voice requires either human interaction or high-quality speech simulation. The external voice experience is the highest-risk surface — caller trust depends entirely on voice quality and conversational naturalness.

QA Approach Options

Three approaches were discussed. The recommended approach is a hybrid of options 2 and 3, with the exact mix determined by BlueWave's QA budget.

Option 1: Internal Testing Only — Mirror Factory and BlueWave team call the system, attempt to break it. Fast and low-cost. Lowest coverage.

Option 2: Hired Human Callers — External callers run scripted and adversarial scenarios. Highest fidelity. Higher cost.

Option 3: ElevenLabs TTS Scripting — Write call scripts, use text-to-speech to simulate callers at scale. Middle-ground: higher coverage than internal, lower fidelity than real humans.

Recommended Approach

A hybrid of Options 2 and 3. The exact ratio depends on budget:

- **Higher budget:** Robust human caller program + TTS automation for regression. Pre-launch target: ~100+ scripted test calls.
- **Lower budget:** TTS-heavy with targeted human testing for critical paths. Pre-launch target: ~50+ scripted test calls.

Both are viable. The plans look very different. Budget is a design input for the QA plan.

Pre-Launch Testing Requirements

Before any live deployment (Phase 1 → Phase 2 transition):

- All three routing paths (Sales, Admin, Candidates) tested end-to-end
- Recruiter Agent screening validated against at least 3 job types
- Edge cases tested: unclear intent, caller frustration, immediate human request, repeat callers, wrong number
- Ticketing validated: creation, progressive fill, merging, contextual summaries
- Messaging bus validated: correct channel, correct person
- Voice quality validated across environments (mobile, landline, speakerphone, background noise)

Success Metrics

Success metrics will be defined during Design & Development (see §14). Candidate metrics:

- Call resolution rate (% resolved without human escalation)
- Time-to-qualified-shortlist (outbound screening)
- Recruiter hours saved per week
- Candidate satisfaction signal (qualitative, from recruiter feedback)
- Ticket accuracy (% rated "useful" by receiving team member)
- Error rate (% of calls requiring post-hoc correction)

---

11. Out of Scope

- **Website chatbot.** Discussed during discovery but designated as an Expansion item, not MVP.
- **Direct sourcing tool integration.** LinkedIn Recruiter, ZipRecruiter, Indeed, and Dice excluded until cost controls are established.
- **Sales tools and compliance tools.** Future possibilities, not part of this engagement.
- **Multi-tenant architecture.** Out of scope until the productization decision is made (Phase 3 gate).
- **ADP integration.** Agent provides self-service instructions from knowledge base only.
- **Go Marketing / Adam coordination.** Not in scope unless Charlie explicitly directs.

- **Training and change management.** Mirror Factory supports onboarding; formal training programs are not scoped deliverables.

12. Constraints & Dependencies

Data Privacy

BlueWave requires a private, closed-loop AI deployment. No candidate data, call recordings, or business data should be exposed externally. Cloud models are not fully air-gapped — this trade-off must be evaluated during Design & Development.

No Internal Engineering Staff

BlueWave has no engineering or IT team. Charlie Lewis and Connor Meehan are the technical POCs. All tooling must be CMS-level simple. Mirror Factory provides ongoing service, not just a code delivery.

Per-Click Sourcing Cost Controls

AI agents must not have uncontrolled access to per-click platforms. Budget guardrails, exclusion rules, and cost caps are hard prerequisites for the outbound screening component.

Maintenance Model

BlueWave expects CMS-like self-service for routine operations with a service agreement for complex changes. This is a hybrid consulting and platform engagement with ongoing support.

Voice QA Limitations

Voice QA cannot be fully automated. The QA approach and budget directly affect pilot quality and timeline.

Client Technical Capacity

Charlie and Connor's availability for whiteboard sessions and design feedback is a direct dependency on the Phase 1 timeline.

Existing Systems

Bullhorn API access level needs confirmation. Internal messaging SOPs are not formalized. Call routing logic has not been documented at the department level. These are design inputs for Phase 1.

13. IP Ownership

Ownership Table

Component	IP Owner	Rights
Aqueduct orchestration layer	BlueWave	Full ownership — source code, documentation, all custom logic
Call routing rules and decision engine	BlueWave	Full ownership
Native ticketing system	BlueWave	Full ownership
Recruiter screening rubrics and logic	BlueWave	Full ownership
Bullhorn and RingCentral integrations	BlueWave	Full ownership (custom to their stack)
Internal messaging bus configuration	BlueWave	Full ownership
Layers platform (context management)	Mirror Factory	Perpetual, irrevocable license granted to BlueWave
Authoring tools	Mirror Factory	Perpetual, irrevocable license granted to BlueWave
Integration framework and connectors	Mirror Factory	Perpetual, irrevocable license granted to BlueWave

Key Principles

- **BlueWave owns everything custom-built for them.** Full source code, full documentation, no lock-in.
- **Mirror Factory retains platform IP (Layers).** This is pre-existing infrastructure that existed before the engagement and will be used across future clients.
- **BlueWave is not locked in.** Aqueduct can function independently of Layers using generic markdown. The license is perpetual and irrevocable.
- **The building analogy:** Layers is the foundation and plumbing — BlueWave doesn't own the concrete mix recipe, but they own their building. They can replace the foundation if they choose.

Open Decision

Whether Aqueduct is purely an internal tool or a licensable product for other staffing firms (NextPath was mentioned) significantly affects architecture decisions. If licensable, multi-tenancy, enhanced documentation, and portability features would be required. This decision should be made before or during Phase 1 (see §14, Question #4).

14. Assumptions & Open Questions

Confirmed Assumptions

Based on the March 31 discovery meeting and April 3 internal discussion. To be validated during Phase 1 kickoff.

- **BlueWave's primary ATS/CRM is Bullhorn** with approximately 200,000 candidate records.
- **BlueWave uses RingCentral** for telephony.
- **BlueWave uses Slack** for internal team communications.
- **Charlie Lewis is the sole strategic decision-maker.** Michael Bowles (CFO) is likely involved in budget approval.
- **Charlie and Connor are the technical POCs.**
- **BlueWave has no internal engineering or IT staff.**
- **Inbound call volume is approximately 8–10 calls per day.**
- **BlueWave has 18–20 recruiters** spending 75–80% of time on unqualified candidates.
- **The website job board syncs to Bullhorn automatically.**

Open Questions Requiring Client Input

Design inputs, not qualification blockers. Must be answered before or during Phase 1.

#	Question	Impact	Owner
1	**Budget range.** Budget determines QA approach (human callers vs. TTS vs. hybrid). Both are viable — the plans look different.	QA plan design, pilot scope	Charlie / Michael Bowles
2	**Full call routing logic.** Who answers today, what do they ask, how do they decide where to route? Document per department.	Core routing engine. Cannot build without this.	Laura / Connor / Kourtney
3	**Bullhorn API access level.** Confirm read/write on: candidate records, tear sheets, job reqs, transcripts, evaluations.	Scope and feasibility of all Bullhorn features	Connor / Charlie
4	**Internal tool vs. licensable product.** Purely internal, or envision licensing to other firms?	Multi-tenancy, portability, documentation. Massive scope difference.	Charlie

5	**Internal messaging SOPs.** When text vs. Slack vs. email? Department-specific rules? Urgency tiers?	Messaging bus configuration. Cannot build without this.	Laura / Kourtney
6	**Top 3–5 qualifying criteria** used across job types for screening.	Defines initial Recruiter Agent logic	Connor / Rachel
7	**Charlie and Connor availability** for whiteboard sessions and design feedback.	Phase 1 timeline. Constrained = longer.	Charlie / Connor
8	**Per-click sourcing costs** for Indeed, Dice, ZipRecruiter. Budget guardrails wanted?	Outbound cost controls. Required before agent access.	Charlie / Connor
9	**Success metrics.** What does "working" look like? What makes the Pilot a clear success or failure?	Pilot evaluation, go/no-go gate	Charlie / Laura
10	**Budget approval process.** Who beyond Charlie approves? Present to CFO or full leadership?	Engagement timeline, presentation planning	Charlie

Internal Mirror Factory Assumptions

- **Pricing model** must be aligned internally before the proposal. Recent deals stalled on pricing communication — must be resolved for BlueWave.
- **Layers IP boundaries** must be formally documented before the engagement begins.
- **Team capacity** for this engagement alongside other active projects. BlueWave is the highest-priority near-term opportunity.
- **First Layers platform client.** Architecture decisions carry forward to all future engagements.

15. Monitoring & Operations

Post-Launch Monitoring Plan

AI systems degrade over time as the world changes beneath them. Unlike traditional software that works until something breaks, AI models and agent behaviors drift. This section defines the

monitoring infrastructure that must be planned during Design & Development and operational during the Pilot.

Agent Performance Monitoring:

- Track routing accuracy (did calls reach the right department?)
- Track screening quality (do recruiter-screened candidates match agent-screened assessments?)
- Track escalation rate (is the agent escalating too much or too little?)
- Track voice quality metrics (latency, dropped calls, caller hang-up rate during agent conversation)

Model/Prompt Performance:

- Monitor LLM response quality over time — detect drift in conversation quality or routing accuracy
- Version all prompts and context profiles; compare performance across versions
- Establish baseline metrics during Pilot; alert on deviations beyond acceptable thresholds

Operational Dashboard:

- Daily summary: calls handled, calls escalated, tickets created, errors flagged
- Weekly review: agent action audit, correction patterns, performance trends
- Accessible to BlueWave's non-technical team (CMS-level simplicity)

Retraining and Improvement Triggers

- Error rate exceeds [TBD]% threshold for any routing path → review and update agent context
- New call type appears that doesn't fit existing routing paths → add routing rule
- Recruiter feedback indicates declining screening quality → review evaluation rubric
- New job types or industries enter BlueWave's portfolio → update screening criteria templates

90-Day Post-Pilot Review

At the conclusion of the Pilot phase, Mirror Factory and BlueWave will conduct a formal review covering: agent performance against baseline metrics, corrections log analysis, system reliability, and recommendations for the Expansion phase. This review informs the go/no-go gate for Phase 3.

---

16. Appendices

A. Reference Documents

Document	Location

Triage Brief	`Triage_BlueWave_2026-03-31.md` (Project context)
Full Intake Document	`Project_Intake_BlueWave_Aqueduct.md` (Project context)
Architecture Diagram v1	`BlueWave_Aqueduct_Architecture.html` (Project context)
Whiteboard Photo	`Bluewave_high_level_MVP_architecture.png` (Project context)
Discovery Meeting Notes	Granola: "BlueWave AI Discussion" (Mar 31, 2026)
Internal Architecture Discussion	Granola: "Discussing BlueWave Project" (Apr 3, 2026)
Pipeline Sync Notes	Granola: "MF Pipeline Sync" (Apr 2, 2026)
Scope Best Practices Research	`Scope_Best_Practices_Research.md` (Project context)
Mirror Factory Pricing Framework	`mirror-factory-pricing-framework_v1.md`

B. Competitive Landscape

Competitor	Capability	Gap
RingCentral AIR	Closest off-the-shelf voice AI for call routing	No Bullhorn integration. Data not portable.
Bullhorn Amplify	Native Bullhorn AI features	No voice AI. Limited to in-platform automation.
Tenzo AI	Candidate screening + Bullhorn integration	Outbound-focused. No multi-caller-type inbound triage.
Whippy / Brilo / Paradox	Various staffing AI point solutions	Point solutions. None bridge inbound voice triage + deep Bullhorn integration.

No existing product handles the full problem. This gap validates the custom build.

C. Glossary

Term	Definition
Aqueduct	The orchestration layer — BlueWave's IP. Routing logic, ticketing, staffing integrations.
Layers	The platform layer — Mirror Factory's IP. Context management, agent profiles, authoring tools.
Background IP	Pre-existing intellectual property owned by Mirror Factory (Layers). Licensed to client.

Foreground IP	Custom IP built for the client during the engagement (Aqueduct). Owned by client.
Tear Sheet	Bullhorn feature — a curated candidate list matched to a specific job requisition.
Progressive Delegation	Starting with low-stakes tasks and expanding AI responsibility as confidence builds.
Context Authoring	Mirror Factory's methodology for structuring the information AI agents use for decisions.
Requisition (Req)	A job order from a client company specifying role, requirements, and placement terms.
JTBD	Jobs-to-be-Done — a framework for defining scope around business outcomes rather than features.

D. Downstream Document Traceability

This scope document feeds two downstream client-facing documents. This table shows how content flows.

Scope Section	ROM Section	Proposal Section
§1 Executive Summary	Project Context (1–3 sentences)	Proposal Overview
§2 Product Vision & Objectives	Implicit context	Goals and Success Criteria
§3 System Architecture	Scope table (Architecture group)	Approach
§4 Agent Architecture	Scope table descriptions	Approach
§5 Feature Specifications	Scope table (workstream descriptions)	Scope of Work
§6 Data Requirements	Key Assumptions	Dependencies
§7 Integration Map	Scope table (Data & Integration group)	Scope of Work, Dependencies
§8 User Experience Surfaces	Scope table descriptions	Approach
§9 Phased Roadmap	Timeline Range	Schedule table
§10 Testing & QA Strategy	Key Assumptions	Approach, Deliverables
§11 Out of Scope	Out of Scope	Out of Scope
§12 Constraints & Dependencies	Key Assumptions	Dependencies, Assumptions
§13 IP Ownership	Not included (per ROM spec)	IP and Licensing
§14 Assumptions & Open Questions	Key Assumptions	Dependencies, Assumptions, Change Control
§15 Monitoring & Operations	Future Scope reference	Support and SLAs

This document is a living specification. It will be updated as open questions are resolved and design decisions are made during Phase 1. Version history is maintained in the document header.

Version History:

Version	Date	Changes
0.1	April 5, 2026	Initial draft — 12-section structure
0.2	April 5, 2026	Revised: restructured executive summary, added JTBD statements, agent architecture section, data requirements section, IP ownership table, monitoring & operations section, MoSCoW prioritization, hypothesis-driven AI scope, voice latency budgets, downstream traceability table, version history